



Fundamentals Of SE

Project Title:

Crime-Guard: Crime Analysis and Prediction System

Project Introduction:

Crime Guard is a desktop application designed to analyze, visualize, and predict crimes and their patterns based on historical crime data. The data will be imported from secure and accurate websites of law enforcement agencies, certain data preprocessing techniques will be applied to clean the data, store that data in relational databases, export them into a data structure for analysis, visualize the data using several statistical techniques and make predictions for user safety by applying advanced supervised and unsupervised machine learning algorithms. Provides with real time crime spikes, anomalies, and red zones. An appealing and user-friendly application interface coupled with powerful features to build a crime app never seen before.

Team Members Names and Roles:

Team Lead	Muhammad Bilal(L22-7551)
Developer	Akhyar Chaudhary(L22-7461)
Tester	Ahmed Abdul Muttal(L22-7466)
Architect	Umair Imran(L22-8370)

Deliverable Number:

This is sprint number3/iteration 3, therefore, the deliverable number is 3.

Date Of Submission:

The date of submission is 5 May,2024.

Advantages:

- User-friendly interface
- High Speed Performance and Efficiency
- Offline Access
- Easy Data Retrieval and Insertion
- Highly Secure

Constraints:

- Region Limitations: The data is limited to Pakistan
- Language: The application is available only in English language
- Platform Dependency: Only operatable on windows
- Scalability: Hard to scale for growing number of users and demand

User Stories:

-- User Story 1(Visitor):

- As a prison warden, I want to be able to manage all the information regarding visitors and their visits in prison.

--Sub User Stories:

- Scheduling new visit easily after entering data into database
- Appealing pie charts to have a visualization regarding visitors and visits
- Registering New Visitor
- Viewing Medical belongings
- Keeping track of all the belongings of visitors before visits

I know I am done when I can handle visitors and visiting schedules.

--User Story 2(Activity Department):

- As the prison manager, I want to be able to see an overview and control all the activities which are to be done by the prisoners.

--Sub User Stories:

- Viewing each activity individually
- Creating new activity
- Registering prisoner to the activity

I know I am done when I can navigate and have an overview of all the activities

--User Story 3(Cell):

- As a prison warden, I want to effectively record and assign prisoners to cells

--Sub User Stories:

- View Particular cell and cell database
- Add prisoner to cell
- Assign Staff
- Create cell

I know I am done when I will be able to assign and keep track of cells and prisoners

--User Story 4(FIR):

- As a police inspector, I want to be able to register and manage all the FIR's given by the citizens.

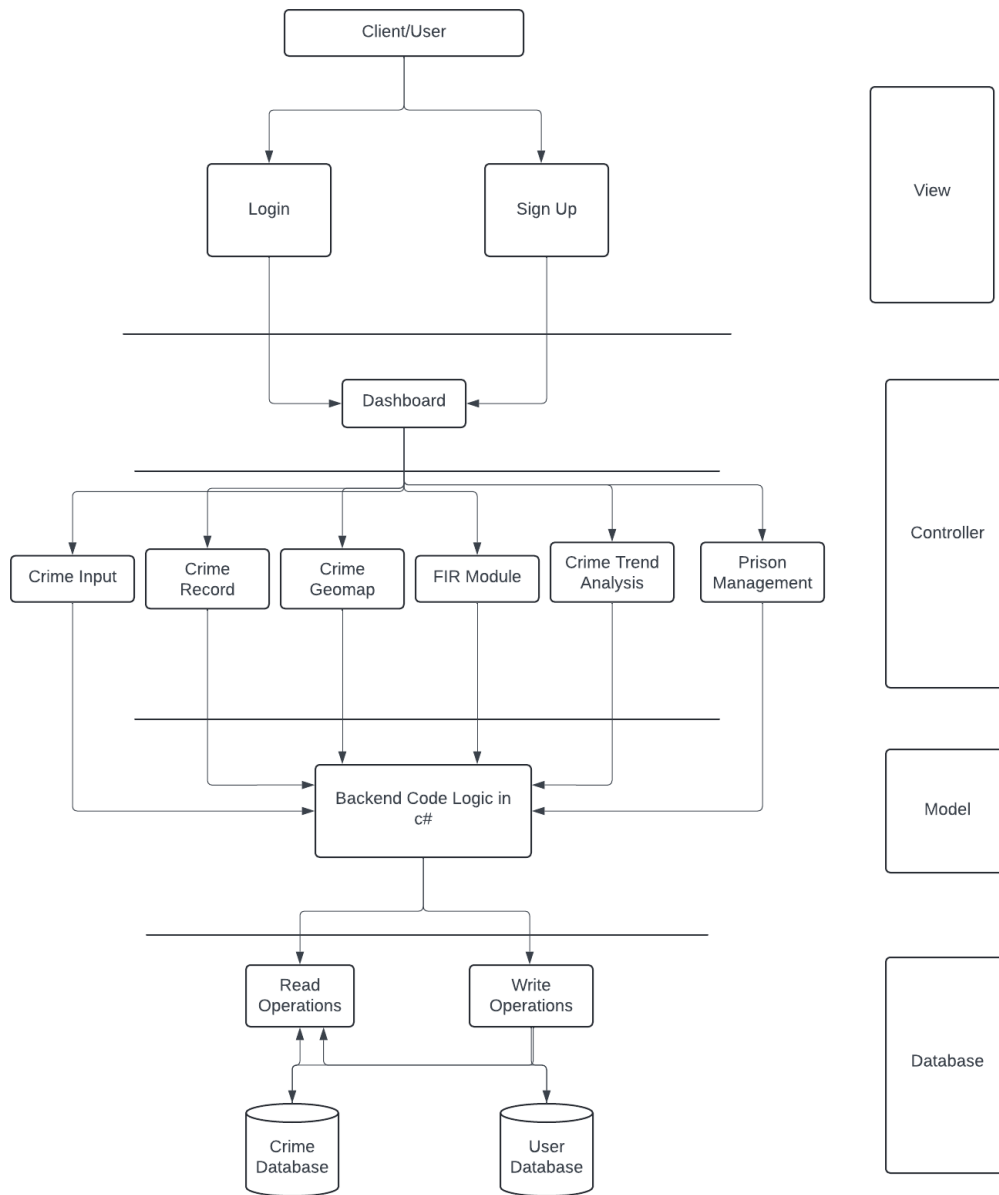
--Sub User Stories:

- Register New FIR
- View FIR Records
- Update FIR
- Delete FIR

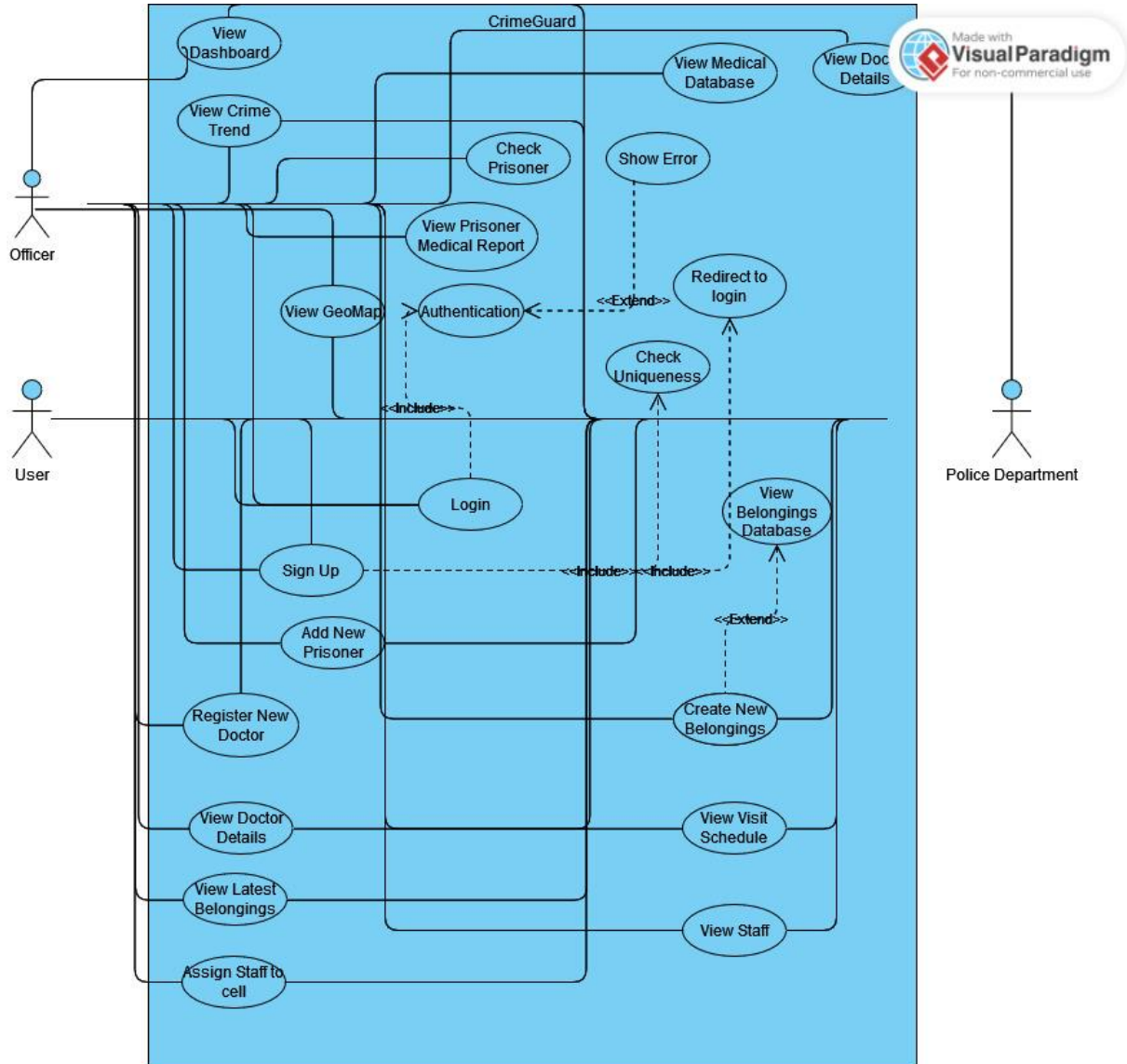
I know I will be done when I am able to register and keep track of all the FIR's.

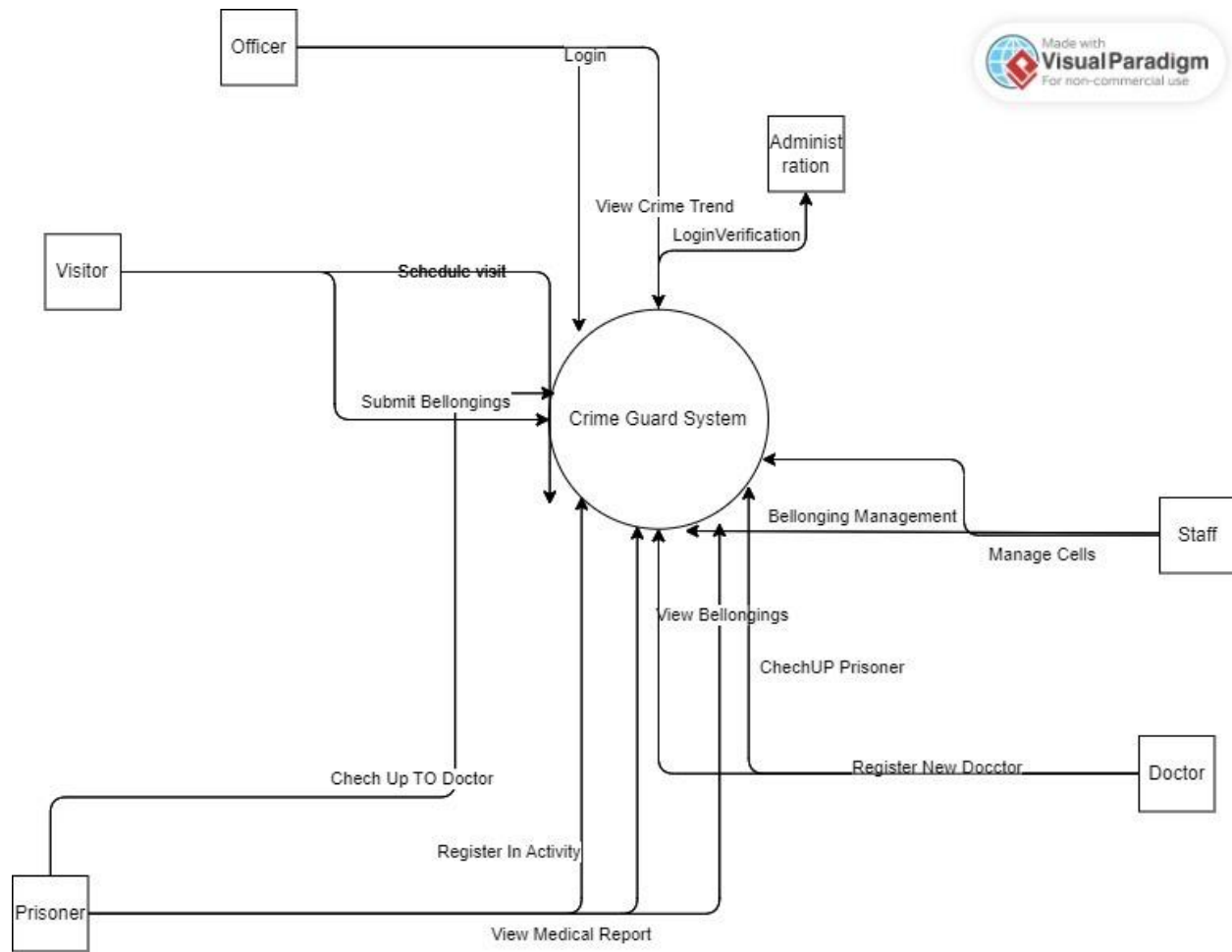
Architecture:

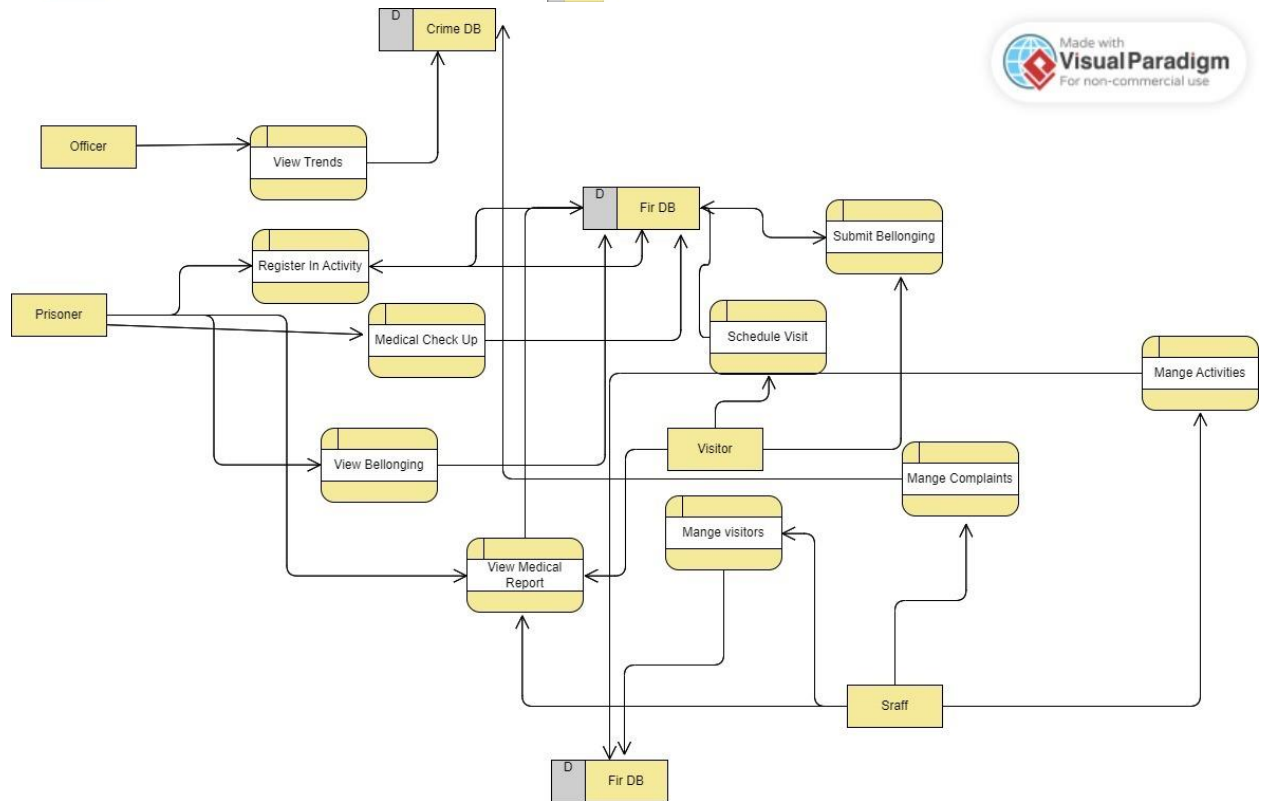
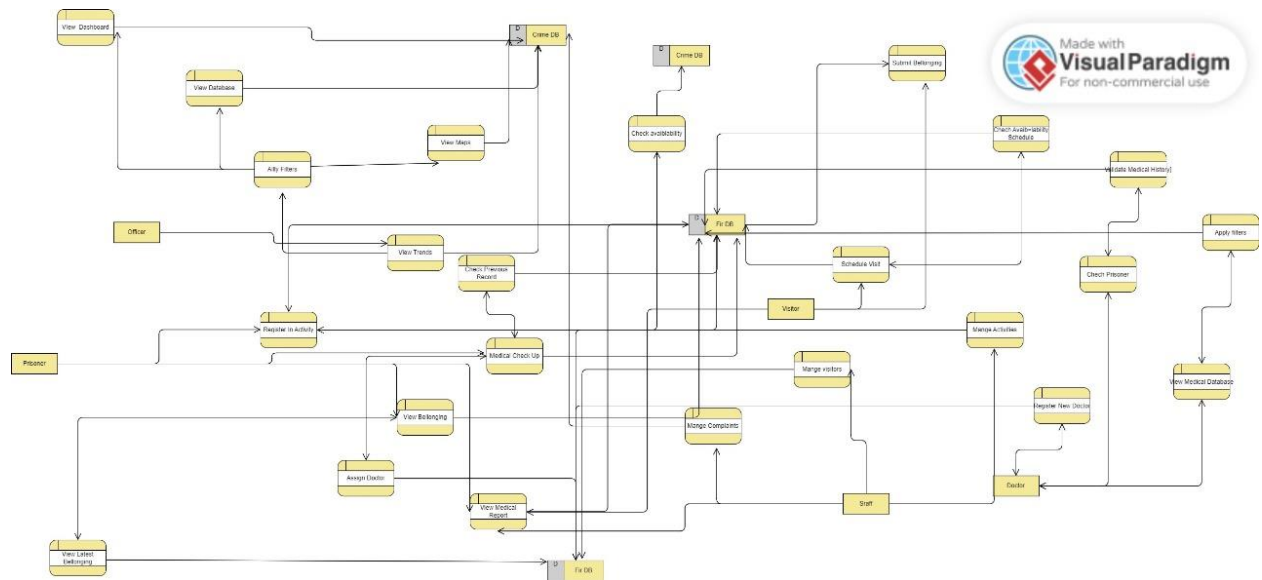
The most suitable architecture for our CrimeGuard application is MVC+Layered.

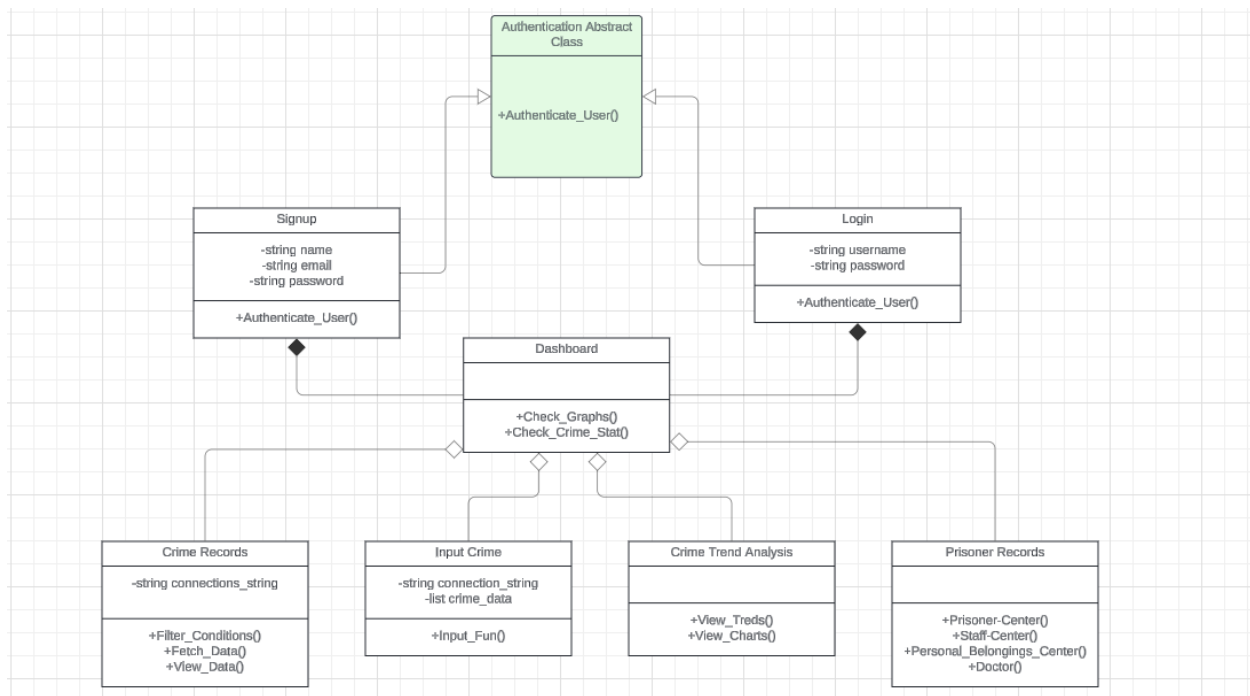
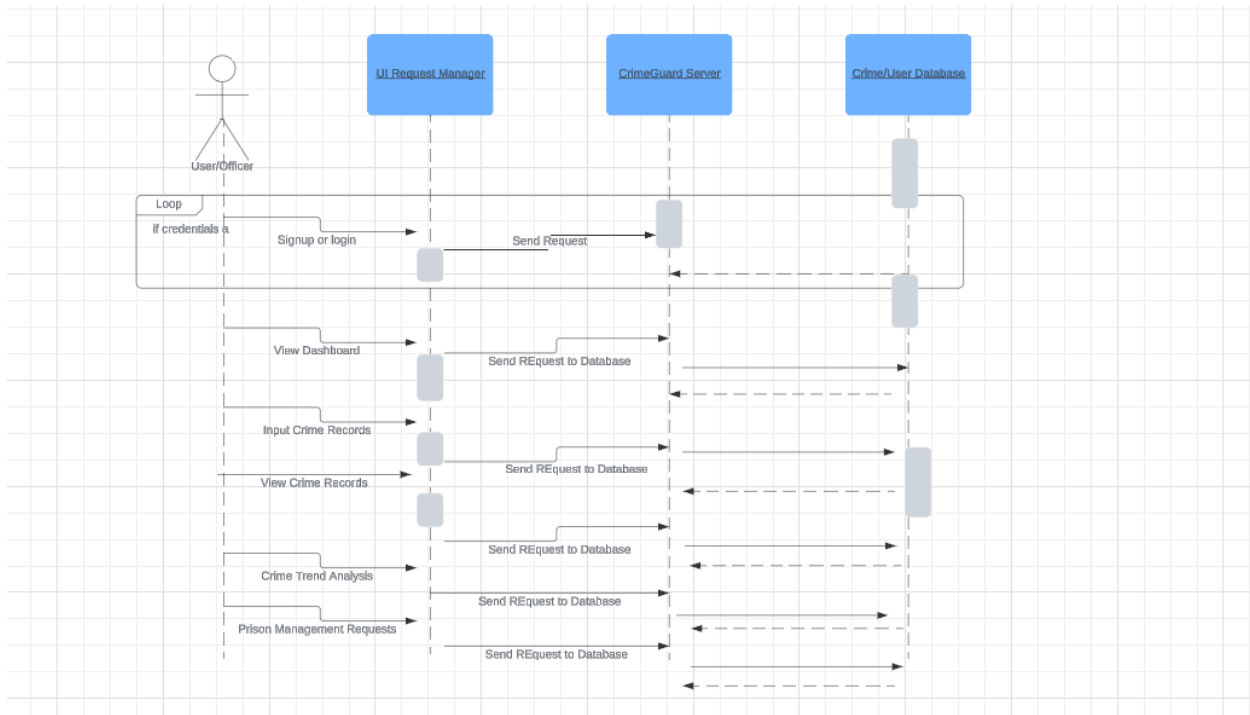


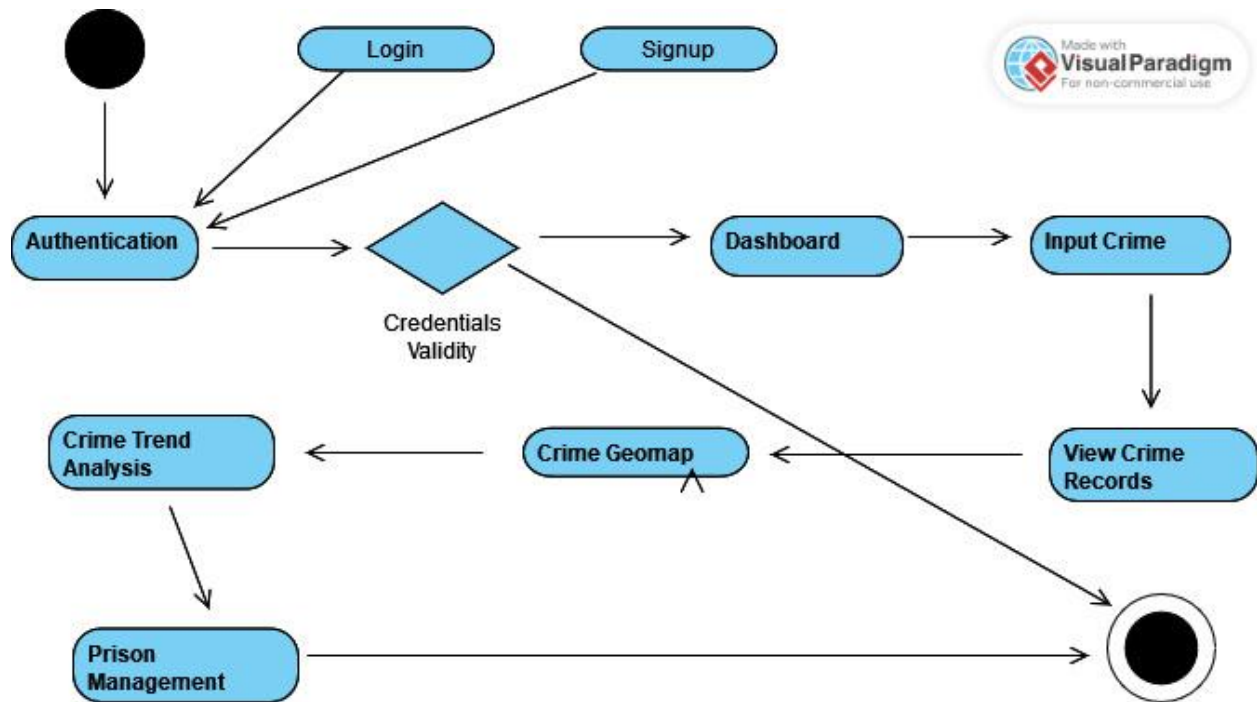
Design (Sprint 2 Items):



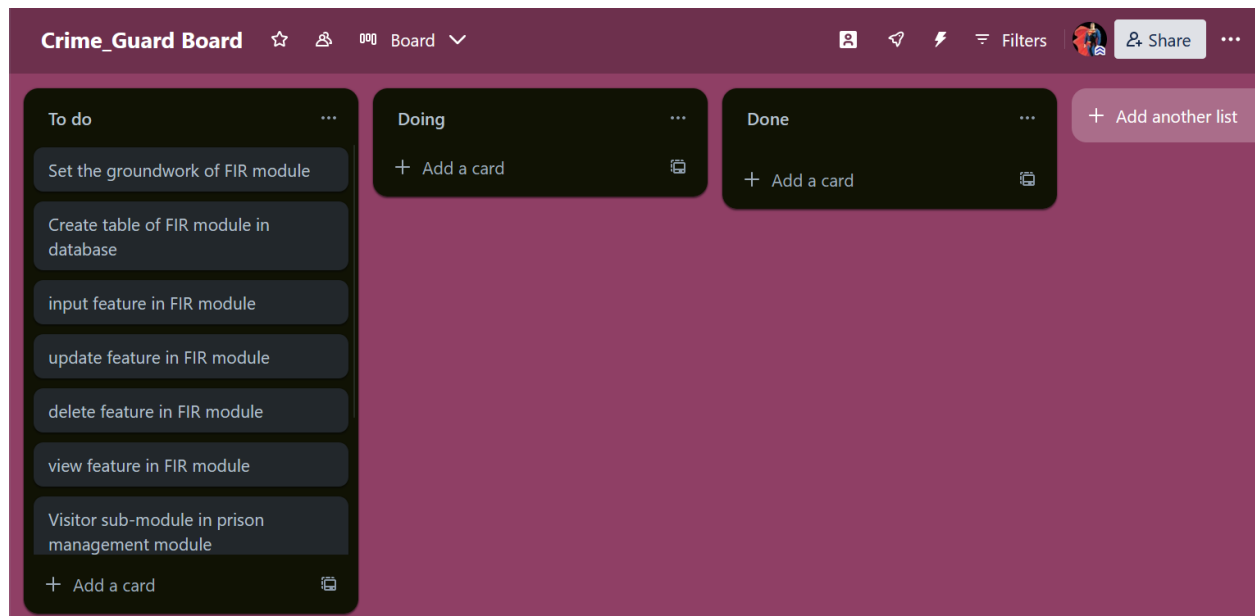








Scrum Board:



Crime_Guard Board

Board

Filters

Share

To do

update feature in FIR module

delete feature in FIR module

view feature in FIR module

Activity Department feature in prison management module

Cell Department feature in prison management module

+ Add a card

Doing

Visitor sub-module in prison management module

Set the groundwork of FIR module

Create table of FIR module in database

input feature in FIR module

+ Add a card

Done

+ Add a card

+ Add another list

work-of-fir-module

Crime_Guard Board

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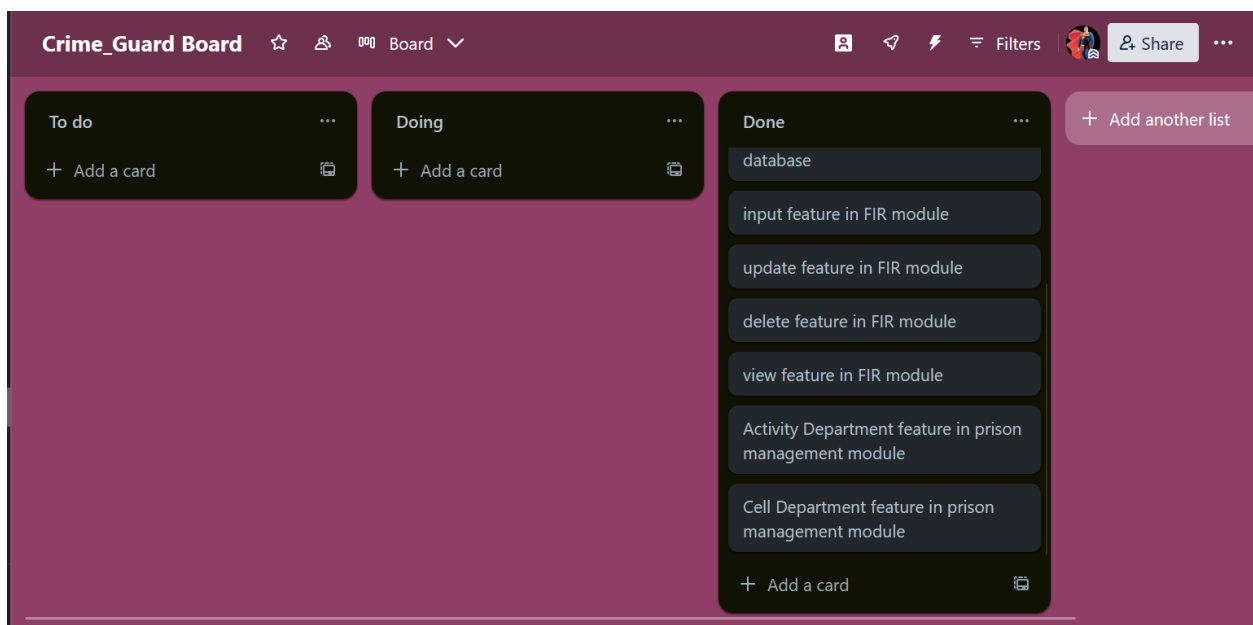
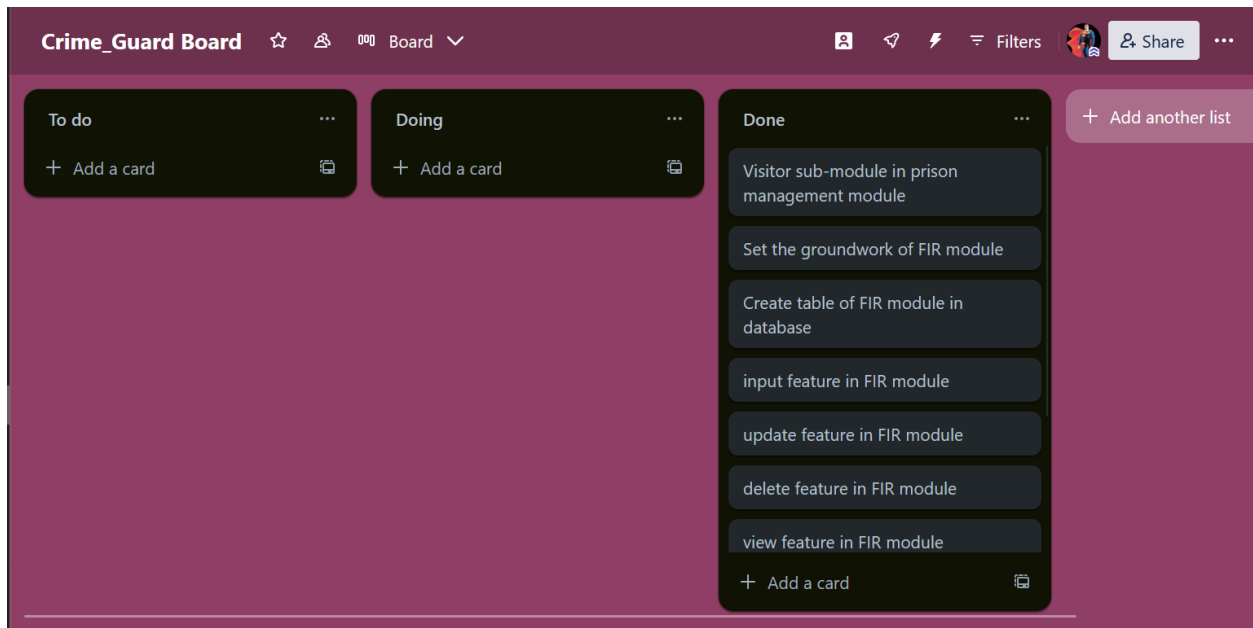
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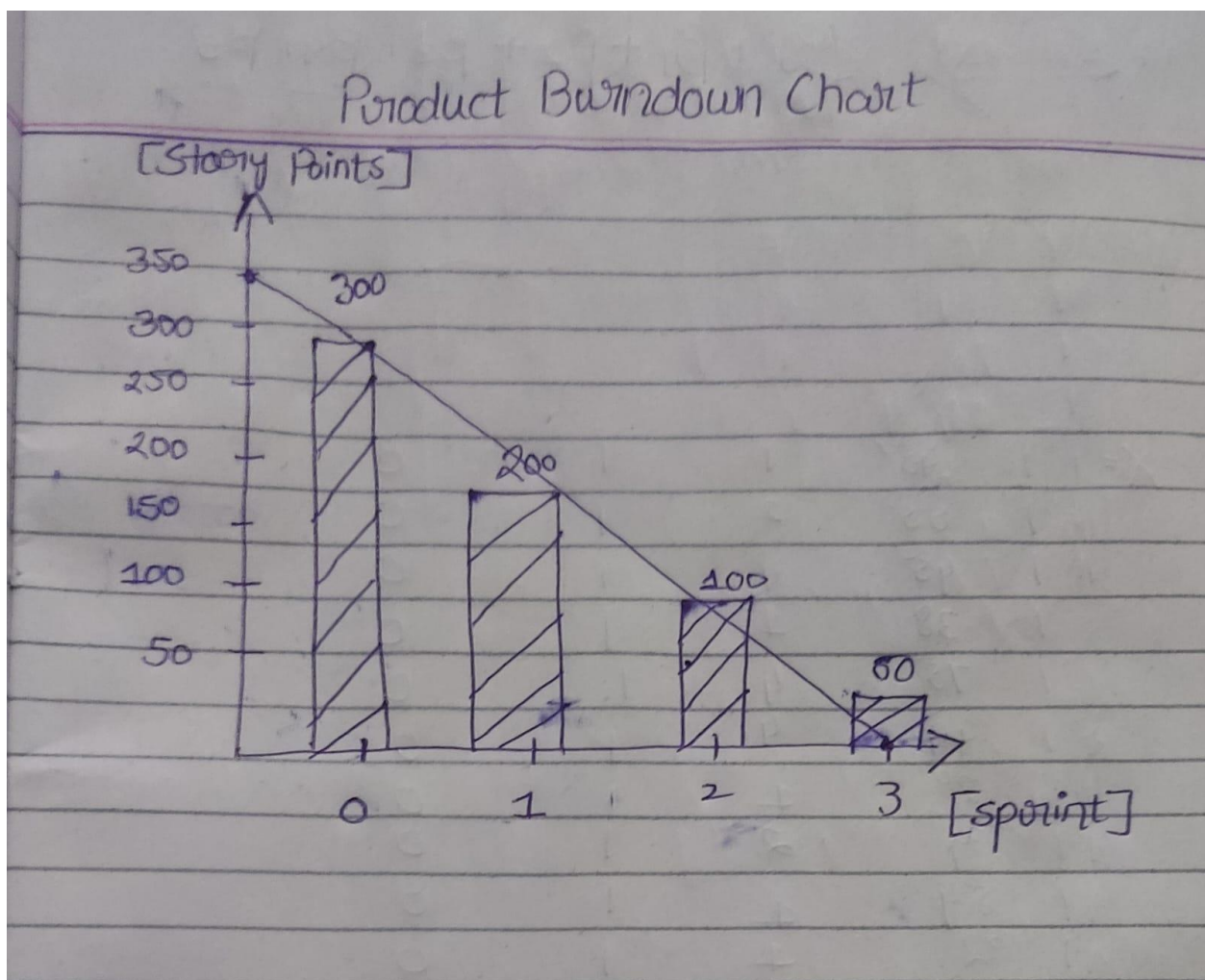
Tik4NTY4?hl=en-GB



NFR (Non-Functional Requirements):

- Usability: The design shall be user-friendly and intuitive for better usability on the user end.
- Security: User Database ensures the safety of user credentials by using industry standard encryption. Regular audits are to be conducted for user and crime databases.

Product Burndown Chart:



Boundary Value Analysis (BVA):

CrimeGuard

Boundary Value Analysis (FIR module CNIC length limit)

Invalid (min-1)	Valid (min, min+1, nominal, max-1, max)	Invalid (max+1)
12	13, 14, 14, 14, 15	16

Test Case	Input	Output
1	35200-21546321	Valid
2	35202-2810498-2	Valid
3	3-5202-2810498-2	Invalid
4	351001424611	Invalid
5	352002154631-9	Valid

Iteration 2 Implementation Screenshots:

Prison Management System

Prisoner Center

Enter

Staff Center

Enter

Personal Belongings Center

Enter

Doctor

Enter

Visitor

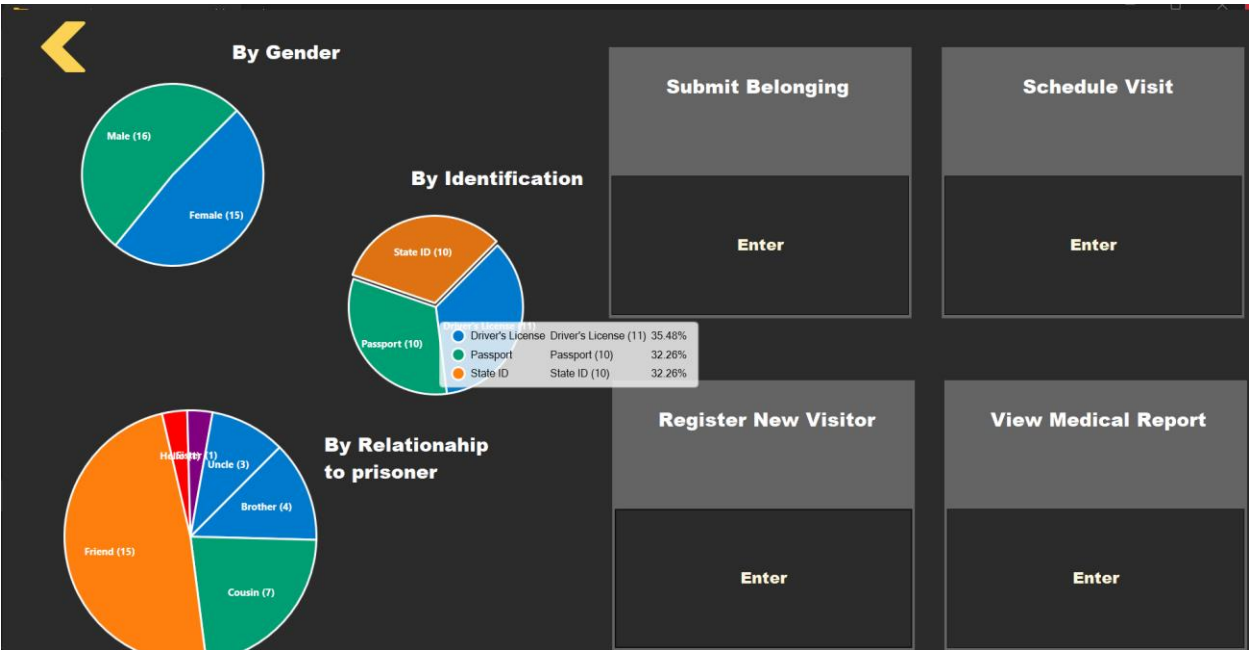
Enter

Activity Department

Enter

Cell Department

Enter



Next

	complainNo	phoneNumbr	name	fatherName	cric	userEmail	age	gender	province	homeDistrict	homePoliceS	address	id	dateOfIncide	place	d
1	1	John Doe	Michael Doe	12345-678	johndoe@exa	30	Male	Punjab	Lahore	Model Town	123 Main	1	1/5/2023	Main Street	L	
2	2	Jane Smith	David Smith	98765-432	jane@exa	25	Female	Sindh	Karachi	Clifton	456 Park A	2	2/10/2023	Shopping	K	
3	11122233	Alice John	Bob Johns	11122-233	alice@exa	35	Female	Khyber Pa	Peshawar	City Center	789 Elm St	3	3/10/2023	University	P	
4	03025022	John Doe	Michael Doe	12345-678	johndoe@exa	30	Male	Punjab	Lahore	Model Town	123 Main	4	4/5/2023	Market Area	Q	
5	03001234	John Doe	Michael Doe	12345-678	johndoe@exa	35	Male	Punjab	Lahore	Model Town	123 ABC	5	5/18/2023	Residential	Is	
6	03123456	Jane Smith	David Smith	23456-789	janesmith	28	Female	Sindh	Karachi	Clifton	456 XYZ	6	6/22/2023	Public R	Rs	
8	03345678	Michael Br	James Bro	45678-901	michaebro	30	Male	Balochistan	Quetta	Can'tt	101 EFG L	8	8/14/2023	Hospital	H	
9	03456789	Emily Wilson	Daniel Wils	56789-012	emilywilson	25	Female	Gilgit-Balt	Gilgit	City	202 HJI St	9	9/5/2023	Constructi	A	
10	03567890	Matthew L	Joseph Le	67890-123	matthewle	38	Male	Azad Jam	Muzaffara	Gharibabad	303 KLM A	10	10/17/2023	Bank Bran	G	
11	03678901	Emma Gar	Christophe	78901-234	emmagar	33	Female	Punjab	Faisalabad	Gulberg	404 MNO	11	11/20/2023	Cinema Ha	L	
12	03789012	William Ma	Richard M	89012-345	williammart	45	Male	Sindh	Hyderabad	City Center	505 PQR	12	12/25/2023	Restaurant	M	
13	03890123	Olivia Robi	Thomas R	90123-456	oliviabobins	31	Female	Khyber Pa	Abbottabad	Jinnah Road	606 STU A	13	1/8/2024	Bus Stop	S	
14	03901234	Daniel Hall	George Hall	01234-567	danielhall	40	Male	Balochistan	Gwadar	Fish Harbor	707 VWX	14	2/13/2024	Railway St	M	
15	04012345	Sophia Lo	Kevin Lopez	12345-678	sophialope	27	Female	Gilgit-Balt	Sophrulo	Bazaar Ro	808 XYZ	15	3/17/2024	Public Lib	H	
16	04123456	Alexander	Edward Yo	23456-789	alexandery	36	Male	Azad Jam	Mirpur	Mirpur City	909 ABC	16	4/21/2024	Petrol Pump	M	
17	04234567	Isabella H	Peter Hill	34567-890	isabellahill	29	Female	Punjab	Rawalpindi	Sadar	1010 EFG	17	5/26/2024	Airport	B	
18	04345678	Ryan Allen	John Allen	45678-901	ryanallen	34	Male	Sindh	Sukkur	Sukkur City	1111 HIL	18	6/29/2024	Hotel	G	
19	04456789	Mia Scott	Ryan Scott	56789-012	miascott@	32	Female	Khyber Pa	Nowshera	Nowshera	1212 KLM	19	7/3/2024	Playground	S	
20	04567890	James Gre	Michael Gr	67890-123	jamesgree	37	Male	Balochistan	Khuzdar	Main Bazaar	1313 MNO	20	8/9/2024	Amuseme	S	
21	04678901	Ava Perez	William Per	78901-234	avaperez	26	Female	Gilgit-Balt	Chilas	City Center	1414 PQR	21	9/14/2024	Supermark	S	

Work Division:

1. Module 1(Prison Management Remaining sub-modules (Visitor, Activity and Cell Department)) = Muhammad Bilal and Umair Imran
2. Module 2(FIR Module) = Akhyar Chaudhary and Ahmed Muttal

Lesson Learnt by Group:

1. One of the most significant lessons we learnt was how crucial it was to understand and document the requirements before diving into the development process.
2. We gained valuable insights throughout the development process about how important it was to implement a thoughtful user-friendly design and choose an appropriate architecture.
3. Effective collaboration and communication among the project members were the reasons we always completed the sprints before time in each iteration.
4. We learned the importance of implementing comprehensive testing strategies to identify and address bugs early in the development lifecycle.